

Pre-A Q3LOCK Gibbs, Feshbach, TFIM, and Round-1 Map Fingerprint Checkpoint

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This single gate-level checkpoint synthesizes EXP-000809 and EXP-000810. It advances reusable result R-167 v2.0 and reusable result R-168 v1.1 without changing a claim card or tier. Both results are T0, additive, and `claim_bearing: false`. Their stable result identifiers are

1. [PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#);
2. [PA-ROUND1-PROSPECTIVE-HOLDOUT-FREEZE-PROTOCOL-AND-CURRENT-TREE-READINESS-AUDIT](#).

R-167 v2.0 proves four narrow children. In finite volume it derives two oppositely oriented Gibbs-weighted unitary cutoff bounds and trace-state stability from one static second moment. It reduces a uniform fixed-edge restricted tail to one growing cubic corridor. It gives a finite-volume, absolute-energy Feshbach and relative-form precursor. Finally, it applies a two-phase quantum Pirogov–Sinai theorem to the exact *compressed* finite-spin transverse-field Ising model. These results do not provide the missing oscillator transfer.

R-168 v1.1 proves two narrow children. The exact current M1-v0, M2-v0, and M5-v0 candidate versions have no admitted map, and adding only a map while preserving all non-map data leaves zero all-PASS survivors. Independently, the finite-torus Gaussian M2 kernel has an exact ordered 48-component node-axis fingerprint, every component equal to one. No physical response map or candidate is created.

The exact historical v0.8 combined source/PDF is prior evidence only for R-167 v1.9 and R-168 v1.0. It is not current evidence for R-167 v2.0 or R-168 v1.1 and cannot satisfy this checkpoint. No per-lemma or intermediate PDF was issued. This v0.9 fragment is the one combined synthesis source for the present logical checkpoint; its sibling PDF may be issued only after all proof, formal-authority, integrated, source-form, freshness, and later render-review gates pass.

2 Closed children, negatives, and open gates

The four R-167 children closed by EXP-000809 are exactly

1. [PA-CP1-ST8-Q3LOCK-FULL-HAMILTONIAN-TWO-ORIENTATION-STATIC-GIBBS-CUTOFF-UNITARY-RESUMMATION](#);
2. [PA-CP1-ST8-Q3LOCK-FIXED-BOND-RESTRICTED-TAIL-TO-GROWING-CORRIDOR-REDUCTION](#);
3. [PA-CP1-ST8-Q3LOCK-BELOW-ONE-HIGH-MODE-FESHBACH-AND-RELATIVE-FORM-SMALLNESS-PRECURSOR](#);

4. PA-CP1-ST8-Q3LOCK-EXACT-COMPRESSED-TFIM-TWO-PHASE-QPS-AND-PHASEWISE-GAP.

Its three new scoped negative results are exactly

1. NG-2026-08-11-PRE-A-ST8-Q3LOCK-WEIGHTED-UNITARY-CUTOFF-AUTOMATIC-ARBITRARY-CONTEXT-AUTOMORPHISM-L2-UPGRADE;
2. NG-2026-08-11-PRE-A-ST8-Q3LOCK-EXTENSIVE-FESHBACH-SELF-ENERGY-AUTOMATIC-QPS-LOCALITY;
3. NG-2026-08-11-PRE-A-ST8-Q3LOCK-STATIC-GAUSSIAN-SYMMETRY-FINITE-MOMENT-AUTOMATIC-FIXED-EDGE-HISTORY-TAIL.

The two R-168 children closed by EXP-000810 are exactly

1. PA-ROUND1-CURRENT-VERSION-M1-M2-M5-MAP-ONLY-ADMISSION-EMPTY-SET;
2. PA-M2-CI8-FINITE-TORUS-GAUSSIAN-DISPERSION-FINGERPRINT.

Its new scoped negative is exactly

NG-2026-08-11-PRE-A-ROUND1-CURRENT-VERSION-MAP-ONLY-ADMISSION-REPAIR.

The new M2 successor gate is

PA-M2-CI8-PHYSICAL-RESPONSE-CHANNEL-AND-ERROR-BOUND.

The following mathematical and prospective parents remain open:

1. PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA;
2. PA-CP1-ST8-Q3LOCK-INFINITE-DIMENSIONAL-RANK-TWO-BAND-BLOCK-DIAGONALIZATION-AND-TWO-PHASE-QPS;
3. PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY;
4. PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE;
5. PA-ROUND1-PER-PARAMETER-COMMON-INPUT-LEDGER;
6. PA-ROUND1-INDEPENDENT-CUSTODIAN-OPAQUE-TARGET-COMMITMENT;
7. PA-ROUND1-ADMISSIBLE-MICROSCOPIC-CANDIDATE-MAP-AND-FROZEN-PREDICTION;
8. PA-ROUND1-CRYPTOGRAPHIC-CUSTODIAN-SIGNATURE-AND-REMOTE-FREEZE-VERIFICATION.

3 Finite-Gibbs two-orientation cutoff resummation

Let H and $H_L = H - W_L$ be bounded-below finite-volume Hamiltonians on a common form domain, and set

$$\rho = Z^{-1}e^{-\beta H}, \quad U(t) = e^{-itH/\hbar}, \quad U_L(t) = e^{-itH_L/\hbar}. \quad (3.1)$$

Assume the two strong Duhamel formulas on a common invariant form core and $\rho(W_L^2) < \infty$. Put

$$\epsilon_L(t) = \frac{|t|}{\hbar} \rho(W_L^2)^{1/2}. \quad (3.2)$$

The ordering with the full flow adjacent to $\rho^{1/2}$ on the right gives

$$\|(U(t) - U_L(t))\rho^{1/2}\|_2 \leq \epsilon_L(t), \quad (3.3)$$

and the opposite ordering gives

$$\|\rho^{1/2}(U(t) - U_L(t))\|_2 \leq \epsilon_L(t). \quad (3.4)$$

No commutation of ρ with U_L is used. Factoring the density-matrix difference into Hilbert–Schmidt products yields

$$\|\rho - U_L(t)\rho U_L(t)^*\|_1 \leq 2\epsilon_L(t). \quad (3.5)$$

For a bounded context satisfying the half-modular identity

$$A\rho^{1/2} = \rho^{1/2}C_A, \quad C_A = \alpha_{-i\beta\hbar/2}(A), \quad (3.6)$$

the right observable estimate is

$$\|[\alpha_t(A) - \alpha_t^{(L)}(A)]\rho^{1/2}\|_2 \leq (\|A\| + \|C_A\|)\epsilon_L(t). \quad (3.7)$$

Applying (3.6) separately to A^* gives the left estimate with $\|C_{A^*}\|$. One narrow sufficient class is a finite Bohr-projective decomposition $A = \sum_\omega A_\omega$ with $[H, A_\omega] = \omega A_\omega$, $|\omega| \leq \Omega$, and finite $\sum_\omega \|A_\omega\|$. This is not an arbitrary bounded-observable theorem.

For the registered finite-volume Q3 hard/form cutoff pair, H and H_L are closed form sums on the common quartic form domain, W_L is a real quadratic coordinate multiplier, and the existing finite-volume Gibbs moments control W_L^2 . Spectral or form truncation of W_L , bounded Duhamel, and then S_2 closure realize (3.3)–(3.5). A separately defined smooth clipped-coordinate cutoff requires its own form-convergence audit.

4 Why arbitrary bounded contexts do not follow

For $0 < p < 1/2$, take

$$\rho_p = \text{diag}(1-p, p), \quad H = \frac{\pi\hbar}{t_0}P, \quad H_L = 0, \quad P = |1\rangle\langle 1|, \quad \beta = \frac{t_0}{\pi\hbar} \log \frac{1-p}{p}. \quad (4.1)$$

Then ρ_p is exactly Gibbs for H . At $t = t_0$, both squared weighted unitary errors equal

$$\|(U - U_L)\rho_p^{1/2}\|_2^2 = \|\rho_p^{1/2}(U - U_L)\|_2^2 = 4p, \quad (4.2)$$

while

$$\rho_p(W^2) = \frac{\pi^2 \hbar^2 p}{t_0^2}. \quad (4.3)$$

For $A = \sigma_x$, the sum of the two squared one-sided automorphism errors is exactly 8, but both evolved density matrices equal ρ_p , so their trace distance is exactly zero. The needed half-modular norm diverges like $\sqrt{(1-p)/p}$. Thus weighted-unitary cutoff control and trace-state stability do not automatically upgrade through every bounded context. The finite Bohr-projective class above is not rejected.

5 Fixed-edge tail to a growing cubic corridor

For the induced nearest-neighbour edges E_R of $C_R = [-R, R]^3$, exact counting gives

$$m_R = |E_R| = 6R(2R+1)^2 \leq 54R^3. \quad (5.1)$$

If $W_{R,L} = \sum_{e \in E_R} w_{e,L}$, quadratic-form Cauchy gives

$$W_{R,L}^2 \leq m_R \sum_{e \in E_R} w_{e,L}^2. \quad (5.2)$$

A translate-, history-, and orientation-uniform fixed-edge estimate

$$\sup_{e \in E_R} \{\sigma_+(w_{e,L}^2) + \sigma_-(w_{e,L}^2)\} \leq c^2 K_{\text{edge}}(L) \quad (5.3)$$

therefore implies

$$\sigma_+(W_{R,L}^2) + \sigma_-(W_{R,L}^2) \leq m_R^2 c^2 K_{\text{edge}}(L). \quad (5.4)$$

For the displayed hard-tail fixture,

$$K_{\text{edge}}(L) = 4e^{-L^2}(L^4 + 2L^2 + 2), \quad (5.5)$$

and $L_R = \sqrt{R}$, $c = 1/3$ give the exact bound

$$1296R^6 e^{-R}(R^2 + 2R + 2) \longrightarrow 0. \quad (5.6)$$

These are hard-tail constants for $w_{e,L} = -c(q_x \cdot q_y)1_{\{X_e > L\}}$, not constants for a smooth clipped-coordinate Q_L . On a periodic cubic torus, translations leave three positive-bond direction orbits; cubic rotations are additionally needed to identify them. The actual Q3 onsite-interspersed fixed-edge history estimate (5.3) remains open.

Static Gaussianity does not supply (5.3). Relative to a standard two-site Gaussian, the tilt

$$\frac{d\nu_\kappa}{d\nu_0}(x, y) = \sqrt{1 - \kappa^2} e^{\kappa xy}, \quad \kappa = \frac{3}{4}, \quad (5.7)$$

has all finite moments and the stated endpoint symmetries, but the order-two likelihood integral diverges because $1 - 4\kappa^2 = -5/4 < 0$. This is a homogeneous dimer or two-site implication counterfixture, not full one-site translation invariance on the cubic lattice and not a Q3 dynamics no-go.

6 Finite-volume Feshbach and relative-form precursor

Let $P_\Lambda = P^{\otimes \Lambda}$, $Q_\Lambda = 1 - P_\Lambda$, and let the unshifted positive onsite high-mode operator obey

$$K_\Lambda = \sum_x k_x \geq \Gamma \sum_x Q_x. \quad (6.1)$$

In the finite-volume absolute-energy normalization, before subtracting any extensive low-band scalar, positivity gives

$$QHQ \geq \Gamma Q. \quad (6.2)$$

For real $E < \Gamma$, the exact low-band Feshbach operator is

$$F(E) = PHP - E - PHQ(QHQ - E)^{-1}QHP. \quad (6.3)$$

This is finite-volume absolute-energy algebra, not thermodynamic isolation of a ground band. On an arbitrary finite cubic subgraph a nearest-neighbour edge overlaps, including itself, with at most 11 edges. Equality holds only for bulk edges or sufficiently large periodic tori. Boundary edges can have fewer overlaps.

For $P_{xy} = P_x P_y$ and $Q_{xy} = 1 - P_{xy}$, the diagonal local-high compression is distinct from the full off-diagonal centered residual. With Young parameters $u, t > 0$, its coefficients are

$$\eta_b = 8c(1 + u^{-1})(1 + t^{-1})A_Q, \quad (6.4)$$

$$\nu_b = 16c[(1 + u)m^2 + (1 + u^{-1})(1 + t)a^2], \quad (6.5)$$

and summing six incident high forms gives

$$\zeta = 6(\eta_b + \nu_b/\Gamma). \quad (6.6)$$

At $u = t = 1$,

$$\eta_b = 32cA_Q, \quad \nu_b = 32cm^2 + 64ca^2. \quad (6.7)$$

The retained asymptotic corridor has full one-bond off-block norm $\epsilon = O(N^{-3})$ and diagonal relative-form coefficient $\zeta = O(N^{-2})$. These are nonempty smallness precursors. A global extensive Feshbach self-energy bound does not imply quasi-locality or a two-phase QPS interaction norm.

Indeed, one high vector of energy Γ coupled with amplitude ϵ to M orthogonal low vectors has exact self-energy

$$\Sigma_M(E) = \frac{\epsilon^2}{\Gamma - E} \mathbf{1}\mathbf{1}^T. \quad (6.8)$$

Its norm is extensive in M , but every off-diagonal entry is nonzero. The missing linked-cluster or rank-two Lie-Schwinger locality is substantive.

7 Exact compressed-TFIM two-phase theorem

This theorem concerns only the exact compressed finite-spin Hamiltonian. At zero effective transverse field, a forward star has local block

$$h_*^{(0)} = J \sum_{i=1}^3 (1 - s_0 s_i), \quad (7.1)$$

whose direct sixteen-configuration spectrum is

$$\{0^{\times 2}, (2J)^{\times 6}, (4J)^{\times 6}, (6J)^{\times 2}\}. \quad (7.2)$$

The phase selector is correctly normalized as $u \sum_x (1 - s_x)$. With the registered derivative convention, its split vector is

$$k = (0, -2). \quad (7.3)$$

The transverse perturbation is bounded, finite range, and parity invariant. The applicable two-phase authority is D. A. Yarotskii, *Quantum Pirogov–Sinai theory*, Russian Math. Surveys 61:2 (2006), 371–372, <https://doi.org/10.1070/RM2006v061n02ABEH004323>. The distinct single-phase CMP paper and math-ph/0412040 are not the authority for this conclusion.

There is an existential $\epsilon_Y > 0$ such that the convention-clean condition

$$\frac{|\delta_{\text{eff}}|}{2J} < \epsilon_Y \quad (7.4)$$

gives exactly two pure translation-invariant infinite-lattice compressed TFIM phases, exponential clustering, and a positive GNS Hamiltonian gap in each selected phase. There is no explicit ϵ_Y or N_0 , no certification of the repository’s finite $r = -9$ diagnostic, and no rank-two oscillator elimination or oscillator GNS gap.

8 Current-version map-only admission theorem

The exact current versions M1-v0, M2-v0, and M5-v0 have no admitted candidate-specific microscopic-to-common-observable map. The frozen survival rule is

survives if and only if every D00–D09 hard row is PASS.

After a hypothetical map-only addition, the eight remaining non-map hard-row cells are exactly

1. M1: D01=FAIL and D02=NOT_ADMITTED;
2. M2: D03=NOT_ADMITTED, D05=NOT_ADMITTED, D06=NOT_TESTED, and D08=NOT_ADMITTED;
3. M5: D04=FAIL and D05=FAIL.

Thus the current-version admitted-map cardinality is zero, and any new version adding only a response-map slot while preserving law, state, reference, boundary, regulator, dynamics, compactness, quotients, critical contract, validation contract, and robustness still has zero survivors. Adding inertial, compact, gauge, state, law, boundary, or regulator data is not map-only. It creates a substantive new version that must rerun all ten hard rows. The scoped negative rejects only a map-only promotion of the current preserved parent data.

The missing M2 response slot is also logically underdetermined. For $\kappa(t) = C|t|$, the two target-free completions

$$\mathcal{R}_1(\kappa) = \kappa/\kappa_0, \quad \mathcal{R}_2(\kappa) = (\kappa/\kappa_0)^2 \quad (8.1)$$

give exponents one and two from the same stiffness. At $t = 1/8$, doubling t gives exact ratios two and four. Neither completion is an admitted physical response map or prediction.

9 Exact M2 finite-torus fingerprint

Take

$$K(k) = r + c \sum_{j=1}^3 (q^2 - k_j^2)^2, \quad h = \frac{2\pi}{L}, \quad q = mh, \quad m \in \mathbb{N}. \quad (9.1)$$

For each of the eight sign nodes $s \in \{+1, -1\}^3$, put $k_s = qs$. For axis i , define

$$d_{\pm} = \frac{K(k_s \pm he_i) - K(k_s)}{ch^4}, \quad S = \frac{d_+ + d_-}{2}, \quad A = \frac{d_+ - d_-}{2}. \quad (9.2)$$

Exact expansion gives

$$S = 4m^2 + 1, \quad A = 4s_i m. \quad (9.3)$$

With $\bar{S}_s = (S_{s,1} + S_{s,2} + S_{s,3})/3$, define

$$R_{s,i} = \frac{A}{S} \frac{4m^2 + 1}{4s_i m} = 1, \quad U_{s,i} = \frac{S}{\bar{S}_s} = 1. \quad (9.4)$$

Lexicographic ordering in the eight nodes, then $i = 1, 2, 3$, then (R, U) gives exactly $8 \cdot 3 \cdot 2 = 48$ ordered components, all exactly one. This is candidate-internal Gaussian mathematics, not a physical dispersion prediction.

The hypothetical successor identifier [PA-M2-CI8-RS-DISPERSION-MAP-v1](#) occurs only in a design object with `status=DESIGN_ONLY` and `candidate_created=false`. Its map, prediction, target, freeze, tag, score, selection, manifest, and hash remain semantically `NOT_CREATED`; this status is not an unresolved draft placeholder. R-168 v1.1 creates no actual candidate or map.

The open response gate requires a target-blind physical channel and a frozen error bound of the form

$$|O_{\text{phys}} - \mathcal{R}_{M2}| \leq \epsilon_{\text{torus}} + \epsilon_{\text{regulator}} + \epsilon_{\text{nonlinear}} + \epsilon_{\text{loop}} + \epsilon_{\text{state/ref}} + \epsilon_{\text{estimator}}, \quad (9.5)$$

with every term computed from frozen inputs and the total strictly below the frozen acceptance margin before disclosure. The fingerprint supplies none of these physical identifications or errors.

10 Prospective-freeze boundary

R-168 v1.1 retains the v1.0 fail-closed schema, its 28 hostile schema classes, and its seven stable current-checkpoint blockers. It does not create or verify an external custodian commitment, target, candidate map, prediction, public remote commit, annotated tag, authenticated receipt, score, or selection. A real prospective freeze still requires the temporal order

$$t_{\text{custodian}} \leq t_{\text{public freeze}} < t_{\text{disclosure}} \leq t_{\text{score}}, \quad (10.1)$$

an externally keyed and signed canonical commitment, at least one admitted microscopic candidate map with a nonempty frozen physical prediction, a complete common-input ledger, and independent cryptographic remote-object and remote-ref verification. The current-version no-go is not a no-go for a substantively new candidate prepared before disclosure.

11 Devil’s-advocate

1. **DISMISSED – Duhamel orientation:** equations (3.3) and (3.4) use opposite orderings. Only the adjacent full flow is commuted through the Gibbs square root.
2. **UPHELD as an open condition – observable context:** trace-state stability does not control arbitrary automorphisms. The exact two-level fixture leaves a divergent half-modular cost visible.
3. **UPHELD as an open condition – covariance:** torus translations leave three edge directions. The actual partial-history covariance and fixed-edge Q3 bound are not supplied.
4. **DISMISSED – overlap factor:** ten counts the other incident edges; the overlap Cauchy bound includes the edge itself and is at most eleven.
5. **UPHELD as an open condition – locality:** an extensive global self-energy estimate is not a connected-cluster or QPS norm.
6. **DISMISSED – theorem source:** the compressed two-phase theorem uses the cited Yarotskii RMS note and the enumerated forward-star data, not the single-phase CMP theorem.
7. **UPHELD as a finite-parameter boundary:** the smallness radius is existential. No displayed finite N or $r = -9$ point is certified.
8. **UPHELD as an oscillator boundary:** a phasewise gap in the exact compressed spin system is not an oscillator gap.
9. **DISMISSED – map-only arithmetic:** all eight residual cells are read from the frozen hard-row matrix, and the all-PASS rule is unchanged.
10. **UPHELD as a physical-map boundary:** the exact 48-component fingerprint does not choose a response channel, units map, state/reference, or error budget.
11. **DISMISSED – prediction status:** the M2-v1 object is explicitly design-only and creates no map, target, prediction, or freeze.
12. **UPHELD as the final scope boundary:** six closed children do not close the common-alpha, oscillator, prospective, Sector-A, or Pre-A parents.

Independent adversarial review is invited especially on the unbounded Duhamel domain passage, disjoint-edge orthogonality in the Feshbach estimate, the precise two-phase theorem hypotheses, the frozen map-only scope, and the physical interpretation firewall for the M2 fingerprint.

12 Reproduction and evidence contract

The R-167 v2.0 primary, non-importing independent, and integrated scripts are

1. [codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split.py](#);
2. [codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_independent.py](#);
3. [codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_verify.py](#).

The R-168 v1.1 primary, non-importing independent, and integrated scripts are

1. `codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol.py`;
2. `codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol_independent.py`;
3. `codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol_verify.py`.

From the repository root, run

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$r167 = (
    'codes\foundations\pre_a_cp1_st8_q3lock_' +
    'local_measured_renyi_semiclassical_doublet_route_split'
)
& $py -X utf8 ($r167 + '.py')
& $py -X utf8 ($r167 + '_independent.py')
& $py -X utf8 ($r167 + '_verify.py')
$r168 = (
    'codes\foundations\pre_a_round1_' +
    'prospective_holdout_freeze_protocol'
)
& $py -X utf8 ($r168 + '.py')
& $py -X utf8 ($r168 + '_independent.py')
& $py -X utf8 ($r168 + '_verify.py')
```

The current proof-layer outputs are R-167 primary PASS 153/153 and independent PASS 117/117, and R-168 primary PASS 205/205 and independent PASS 223/223. After the single combined checkpoint source/PDF, exact shared manifest metadata, formal authorities, stored-result freshness, generated surfaces, dual PDF extraction, and render review all pass, the expected integrated outputs are R-167 PASS 220/220 and R-168 PASS 262/262. These integrated totals are conditional targets until the later PDF and freshness checks complete; a failed check invalidates the total rather than weakening the gate.

No PDF is built by this source-drafting step. No per-lemma or intermediate PDF belongs to R-167 v2.0 or R-168 v1.1. The eventual v0.9 PDF, if all gates pass, is the one current gate-level checkpoint; the historical v0.8 PDF remains strictly historical evidence and is not reused as current evidence.

13 Exact no-overclaim and dependency boundary

The dynamics parent is **OPEN**. The finite-Gibbs estimate and fixed-edge corridor reduction do not prove the actual translate-, history-, source-, mesh-, orientation-, and exhaustion-uniform Q3 tail, the $n \rightarrow \infty$ split limit, or the actual Q3 common alpha.

The rank-two parent is **OPEN**. The absolute-energy Feshbach estimate does not give quasi-local rank-two oscillator elimination, a rank-two oscillator QPS interaction, cutoff removal, or beta-infinity oscillator phase identification.

The spectral parent is **OPEN**. The compressed-spin phasewise gap is not an oscillator GNS gap, an observed temporal mass, or a physical mass gap.

The M2 response parent is **OPEN**. The exact Gaussian fingerprint is not a physical response map, observable prediction, error enclosure, or validation result.

The prospective parent is **OPEN**. No real prospective freeze, external target commitment, admitted candidate prediction, public remote anchor, authenticated receipt, or score exists.

Nothing in this checkpoint proves regulator removal, continuum control, same-Hamiltonian physical-empty comparison, physical vacuum selection, C6, CP1, physical Sector A, or Pre-A closure. The active A5-SECTOR-A-SYNTHESIS remains conditional at its registered scope.

14 Result footer

Result ID: PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT; PA-ROUND1-PROSPECTIVE-HOLDOUT-FREEZE-PROTOCOL-AND-CURRENT-TREE-READINESS-AUDIT.

Reusable results: R-167 v2.0; R-168 v1.1.

Explorations: EXP-000809; EXP-000810.

Precise statement: Finite-Gibbs two-orientation cutoff resummation; fixed-edge corridor reduction; finite-volume Feshbach/relative-form precursor; exact compressed-TFIM two-phase theorem; current-version map-only empty set and exact 48-component M2 fingerprint.

Claim context: C6-SPACETIME-SIGNATURE; claim_bearing false.

Closed children: Four R-167 and two R-168 children, exactly the six identifiers listed in Section 2.

Negative results: Three R-167 and one R-168 scoped negatives, exactly the four identifiers listed in Section 2.

New open gate: PA-M2-CI8-PHYSICAL-RESPONSE-CHANNEL-AND-ERROR-BOUND.

Scope: T0 additive, claim-nonbearing exact/analytic result at finite-volume, compressed-spin, and finite-torus Gaussian scope; no oscillator or physical-map transfer.

Dependencies: R-167 v1.9; R-168 v1.0; EXP-000806, EXP-000807, EXP-000808; registered finite-Gibbs moments, low-band compression, and frozen ten-row survival rule; open fixed-edge, oscillator, custodian, and remote inputs.

Evidence grade: ANALYTIC + EXACT + primary + non-importing independent; integrated evidence remains conditional on the single final checkpoint lifecycle.

Reproduction command: Run the six commands in Section 12 from repository root, primary then independent then integrated.

Expected output: R-167: PASS 153/153, PASS 117/117, PASS 220/220.
R-168: PASS 205/205, PASS 223/223, PASS 262/262.

Falsification gate: Any Duhamel order/context, edge count/tail factor, Feshbach sign/overlap/relative-form coefficient, star spectrum, selector/QPS hypothesis, hard-row cell, fingerprint order, hash, count, provenance, freshness, extraction, or scope mismatch.

Tier before / after: C6 T1 / C6 T1; no claim or tier change.

No-overclaim statement: No actual Q3 common alpha, rank-two oscillator elimination, oscillator GNS gap, physical response map, real prospective freeze, regulator removal, continuum, physical-empty sign, C6, CP1, physical Sector A, or Pre-A closure.

Next required action: Prove the actual fixed-edge Q3 history theorem and quasi-local rank-two oscillator transfer; derive a target-blind physical M2 response with a frozen error budget; obtain and verify a real prospective freeze before disclosure.

PDF policy: No per-lemma or intermediate PDF. Historical v0.8 is prior evidence only; issue one v0.9 checkpoint only after every proof, formal, integrated, PDF, freshness, extraction, and visual gate passes.